

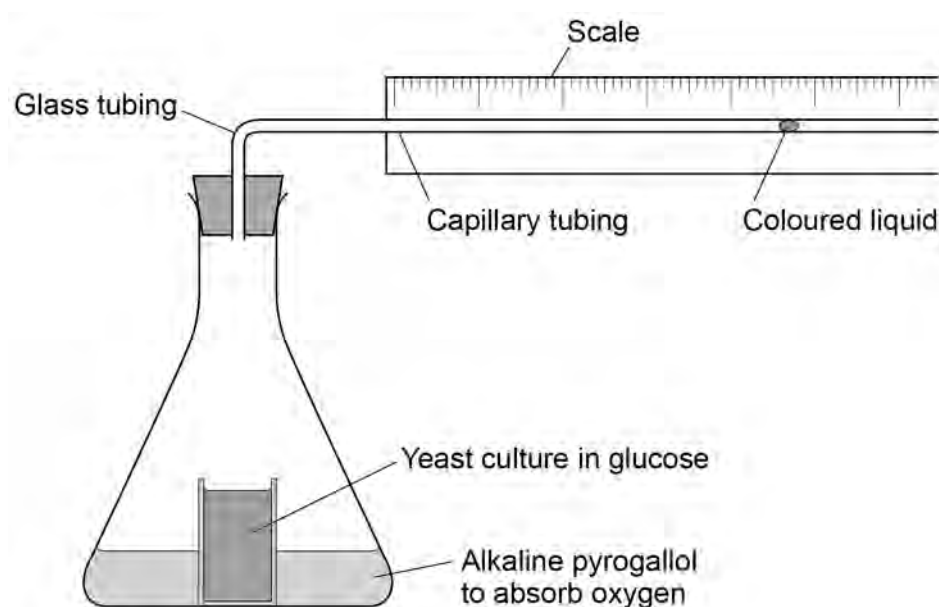
0 3

Yeast cells can respire aerobically or anaerobically. A student used the apparatus shown in **Figure 3** to measure the rate of respiration in yeast.

She:

- positioned the flask in a water bath so that the yeast culture reached a constant temperature
- then left the apparatus for one hour before starting her investigation.

Figure 3



0 3 . 1

Suggest **one** reason why it was important that the student left the apparatus for one hour after the yeast culture reached a constant temperature.

[1 mark]

0	3	.	2
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During her investigation, the coloured liquid moved to the right.

Explain why it moved to the right.

[2 marks]

0	3	.	3
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The student found that the coloured liquid moved 1.5 cm in 24 hours. The diameter of the lumen (hole) of the capillary tubing was 1 mm.

The volume of a capillary tubing is given by $\pi r^2 l$, where π is 3.14 and l = length.

Calculate the volume of gas produced in $\text{cm}^3 \text{ hour}^{-1}$.

Show your working.

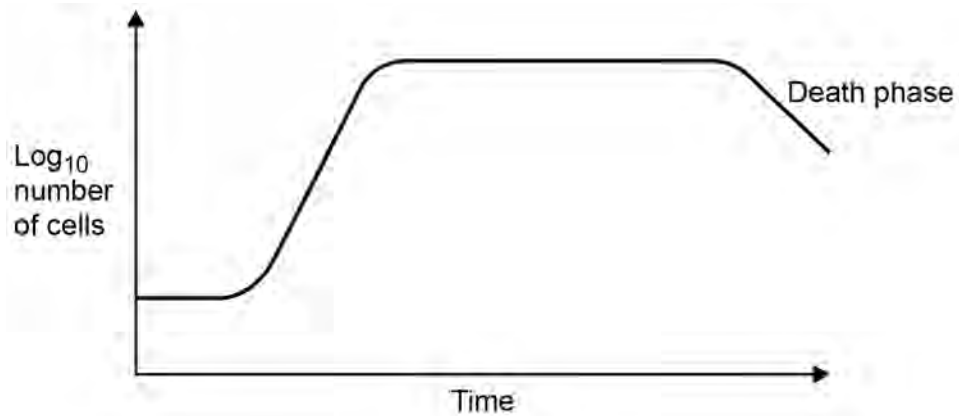
[2 marks]

Answer = _____ $\text{cm}^3 \text{ hour}^{-1}$

Turn over ►

Figure 4 shows a typical population growth curve for yeast under laboratory conditions.

Figure 4



0 3 . 4 Explain why a log scale is used to record the number of cells.

[1 mark]

0 3 . 5 Many yeast cells die during the death phase.

Suggest **one** reason why.

[1 mark]

0	3	.	6
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The following equation can be used to make predictions of the growth in the population of yeast cells under ideal laboratory conditions.

$$X_t = X_0 e^{rt}$$

X_t = the population after a certain time

X_0 = the population at the start

$e = 2.72$ (base of natural logarithm)

r = growth rate

t = time period in hours over which r applies

A population of 2000 yeast cells was left for 10 hours.

The value for the growth rate was 0.5

Assuming no yeast cells died, calculate the predicted size of the population after 10 hours. Show your working.

[2 marks]

Answer = _____

Turn over for the next question

